6 Constraints

First, lets think of an example from last class - the bead on a rotating hoop. If we change to the inertial frame in lock step with the rotation, then we'll have a normal force

$$F_{in} = -N\sin\theta = -m\omega^2 R\sin\theta$$

Then we can fix the Lagrangian expression by appending
$\mathscr{L} = T - U - m\omega^2 R\sin\theta$

Coordinates

If the Hamiltonian is time independent, then
$H = T + U$.

We have the more general expression, for definitionally the Hamiltonian as,

$$\sum_k \frac{\partial \mathscr{L}}{\partial \dot{q}_k}\dot{q}_k - \underbrace{(T-U)}_{\mathscr{L}}$$

$H = T + U$ if

$$\sum_k \frac{\partial \mathscr{L}}{\partial \dot{q}_k}\dot{q}_k = 2T$$

Lets do a transformation for 3d that takes
(x, y, z) as functions of (q_1, q_2, q_3).

Lets define one coordinate system

$$\begin{aligned} r_1(t) = x(t) &= f_1(q_1(t), q_2(t), q_3(t), \cancelto{0}{t}) \\ r_1(t) = y(t) &= f_2(q_1(t), q_2(t), q_3(t), \cancelto{0}{t}) \\ r_1(t) = z(t) &= f_3(q_1(t), q_2(t), q_3(t), \cancelto{0}{t}) \end{aligned}$$

One easy example of this is

$$\begin{aligned} x &= r\sin\theta\cos\phi \\ y &= r\sin\theta\sin\phi \\ z &= r\cos\theta \end{aligned}$$

Our kinetic energy in this new coordinate system is

$$T = \frac{1}{2}(\dot{r}_1^2 + \dot{r_2}^2 + \dot{r_3}^2) = \frac{m}{2}\sum_{n=1}^{3}\dot{r}_n^2$$

We can write that generally

$$\begin{aligned} \dot{r_n} &= \frac{\partial f_n}{\partial q_1}\dot{q}_1 + \frac{\partial f_n}{\partial q_2}\dot{q}_2 + \frac{\partial fn}{\partial q_3}\dot{q}_3 + \cancelto{0}{\frac{\partial fn}{\partial t}} \\ &= \sum_{i=1}^{3}\frac{\partial f_n}{\partial q_i}\dot{q}_i \end{aligned}$$

so

$$T = \frac{m}{2}\sum_{n=1}^{3}\left(\sum_{i=1}^{3}\frac{\partial f_n}{\partial q_i}\dot{q}_i\right)\left(\sum_{i=1}^{3}\frac{\partial f_n}{\partial q_i}\dot{q}_i\right)$$

We can simplify this as long as we keep track of the cross terms

$$\boxed{T = \frac{m}{2} \sum_{nij}^{3} \frac{\partial f_n}{\partial q_i} \frac{\partial f_n}{\partial q_j} \dot{q}_i \dot{q}_j}$$

Remember this, we will come back to it.

This accounts for 27 terms, since for each of the three j's there are three ns, and for each there are three i's.

Remember the overall goal: we're trying to find when

$$\sum_k \frac{\partial T}{\partial \dot{q}_k} \dot{q}_k = 2T$$

We can see that

$$\frac{\partial T}{\partial \dot{q}_k} \dot{q}_k = \frac{\partial}{\partial \dot{q}_k} \left[\frac{m}{2} \sum_{ni} \frac{\partial f_n}{\partial p_i} \frac{\partial f_n}{\partial q_j} \dot{q}_1 \dot{q}_2 \right]$$

We know that our x, y, z were time independent (t is not in the function, and they don't depend on $\dot{q_n}$ (aka holonomic)) so this simplifies to

$$\frac{m}{2} \sum_{nij} \frac{\partial f_n}{\partial q_i} \frac{\partial f_n}{\partial q_j} \frac{\partial}{\partial \dot{q}_k} (\dot{q}_1 \dot{q}_2)$$
$$= \frac{m}{2} \sum_{nij} \frac{\partial f_n}{\partial q_i} \frac{\partial f_n}{\partial q_j} \left[\frac{\partial \dot{q}_i}{\partial \dot{q}_k} \dot{q}_j + \dot{q}_i \frac{\partial \dot{q}_j}{\partial \dot{q}_k} \right]$$

We know that

$$\frac{\partial \dot{q}_i}{\partial \dot{q}_i} = 1 \text{ because ofc}$$
$$\text{and}$$
$$\frac{\partial \dot{q}_i}{\partial \dot{q}_{\text{ not i}}} = 0$$

This looks like a Kronecker delta

$$\frac{\partial \dot{q}_i}{\partial \dot{q}_k} = \begin{cases} 1 \text{ if } i == k \\ 0 \text{ if } i \neq k \end{cases}$$

We can simplify now with

$$\frac{\partial T}{\partial \dot{q}_k} = \frac{m}{2} \sum_{nij} \frac{\partial f_n}{\partial qi} \frac{\partial f_n}{\partial q_j} \left[\delta_{ik} \dot{q}_j + \dot{q}_i \delta_{jk} \right]$$

This simplifies down to

$$\frac{m}{2} \sum_{nj} \dot{q}_j \frac{\partial f_n}{\partial q_j} \left(\sum_i \delta_{ik} \frac{\partial f_n}{\partial q_i} \right) + \frac{m}{2} \sum_{ni} \frac{\partial f_n}{\partial q_i} \dot{q}_i \sum_j \frac{\partial f_n}{\partial q_j} \delta_{jk}$$

Which we can simplify to be

$$\frac{\partial T}{\partial \dot{q}_k} = \frac{m}{2} \sum_{nj} \dot{q}_j \frac{\partial f_n}{\partial q_j} \frac{\partial f_n}{\partial q_k} + \frac{m}{2} \sum_{nj} \dot{q}_j \frac{\partial f_n}{\partial q_j} \frac{\partial f_n}{\partial q_k}$$

so

$$\frac{\partial T}{\partial \dot{q}_k} = m \sum_{nj} \dot{q}_j \frac{\partial f_n}{\partial q_j} \frac{\partial f_n}{\partial q_k}$$

Woah! Lets compare this to the boxed equation above!

$$\sum_k \left(\frac{\partial T}{\partial \dot{q}_k}\right)\dot{q}_k \stackrel{?}{=} 2T$$

$$m\sum_{njk} \frac{\partial f_n}{\partial q_j}\frac{\partial f_n}{\partial q_k}\dot{q}_j\dot{q}_k = 2T$$

Therefore - $H = 2T - (T - U) = T + U$